

Program overview

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Year 2023/2024
 Organization Applied Sciences
 Education Minors Applied Sciences

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Minors TNW 2023							
NB-MI-237 Collaborative Science for Biomedical Breakthroughs 2023	NB-MI-237 Collaborative Science for Biomedical Breakthroughs: Learn Transdisciplinary Skills, Do Research, Explore New Fields						
NB3510	CSBB part 1: Understanding Questions	15					
NB3520	CSBB part 2: developing answers	15					
TN-Mi-189-23	TN-Mi-189 Modern Physics 2023						
TN2305	Quantum Mechanics for the minor	4					
TN2306	Quantum Engineering and Applications	3					
TN2612	Theory of Relativity	3					
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TN2985	Introduction to Methods in Physics	2					
TN2986	Mathematical Methods for Modern Physics	2					
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TN-Mi-219	TN-Mi-219 Quantum Science and Quantum Information 2023						
Minor QSQI for AS/TNW students							
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TN3166	Solid-State Quantum Bits	9					
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TN3156	Quantum Sensing and Measurement	4					
TN3166	Solid-State Quantum Bits	9					
TN3175	Quantum Engineering Group Project	6					
TN-Mi-227-21	TN-Mi-227-21 Sustainable Chemistry and Biotechnology						
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Year	2023/2024
Organization	Applied Sciences
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Minors TNW 2023

Year	2023/2024
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NB-MI-237 Collaborative Science for Biomedical Breakthroughs 2023

Director of Studies	Dr. H. Mira Bontenbal
Minor Coordinator	J.C. Colgrove
In association with the Faculty of	TNW TBM
In association with the University of	Erasmus University Rotterdam and Erasmus MC
EC Program	30 EC
Program Start (Study Year)	2023
Prerequisites	Selection through the minor selection procedure. There are no prerequisites in terms of existing knowledge. We require students to have an interest in science, research, and professional skill development.
Program Structure 1	The minor is split into two blocks of 15 EC. The first block will involve studying the biomedical/societal/nanobiology problem based on a Convergence Flagship (Integrative Neuromedicine), which will be presented by external stakeholders. Stakeholders will indicate what the current status of their field is and what the general problems are. Students will work on these problems during the first block of the Minor, while working on solutions to them during the second block. At the very beginning, students will be assigned into fixed groups of 3-6 students with transdisciplinary intent: ideally, groups should be as heterogeneous as possible in terms of knowledge and background. During this first half of the Minor, students will spend time researching a subtopic of the overarching Minor topic, reporting to each other to exchange knowledge. They will develop a full understanding of resources available for projects, the current state of knowledge and problem, and how to work in a team and who their team is. They will develop the ability to be highly self-reflective and constructively critical.&#8239; Each student group will have a responsible supervisor and a PhD student as coaches. Supervisors will work with and guide weekly each group of students while PhD students will work daily with the student groups to help them with researching the question and developing the project. The responsible supervisor will spend about 4 hours a week with the students, while the daily supervisor will spend a minimum of 5-6 hours per week on this project. A similar structure will take place during the second block of the Minor where students will develop proposals to address the problems in a transdisciplinary fashion. The minor will consist of two courses, NB3510 and NB3520, please see their study guides for complete details of the courses.
Exam requirements	see course descriptions
Administration by the Faculty of	TNW
Minor Coherence / Goal	At the end of the minor students are able to: 1. Communicate research clearly to different audiences and in different settings, both oral, written and visual. a) Create a clear and engaging presentation that is suitable for various audiences (peers, stakeholders and non-academics) in different settings (poster, pitch, symposium). b) Deliver presentations about their research in a clear and engaging way. c) Write an effective research proposal about their research. d) Write an effective scientific report about their research. 2. Collaborate with peers, other groups, stakeholders using transdisciplinary skills to address a challenging question. a) Recognize and describe different roles within (research) groups. b) Assess group processes within their own research group during a project. c) Work effectively in a group which consists of members from different academic disciplines and backgrounds. d) Share findings with other groups in a meaningful way. 3. Develop research skills for understanding a problem and designing projects. a) Find, categorize and integrate different information sources about their research topic. b) Formulate an adequate research question. c) Develop a research strategy based on current knowledge of their topic. d) Develop a project based on current knowledge of their topic. e) Understand ethical considerations and challenges in their research, from lab journals to data management to publication. 4. Reflect on experiences and increase self-awareness in professional environment a) Reflect on experiences in the minor with the provided framework. b) Ask for constructive feedback of superiors and peers. c) Understand how personal biases influence your work and your collaborations. d) Understand the value of making mistakes.
Minor Content 1	Teams of students from multiple disciplines will learn how to approach a research topic and develop a plan for a solution. They will learn what is involved in developing a complete and deep understanding of a research question, then develop a proposed solution (and start working on it). The Minor will be transdisciplinary in essence with strong components of professional skill development and self-reflection. The overarching topic will be one of the Convergence Flagship topics. For 2023-2024, this will be Integrative Neuromedicine. It is both structured and open in terms of challenging students to understand and solve current problems while guiding them in developing the tools to work closely in teams to solve them. Our mission statement: Develop students to work in a transdisciplinary research environment on current biomedical and societal challenges while developing excellence in communication, collaboration, and reflection to be better prepared for the world of tomorrow.
Minor content 2	This minor comes out of the Nanobiology Programme. It will be transdisciplinary, and one important focus will be on how people from very different backgrounds can build a thorough understanding of the issues surrounding the research question. Transdisciplinarity is built into the core of this minor, because we will select Flagship projects and these are transdisciplinary by their very nature, joining researchers, academics, economists, philosophers, clinicians, etc from the three universities participating in Convergence Health & Technology (TUDelft, Erasmus MC and EUR). We will build on this further by developing skills in collaboration, reflection, and integration.

	It will teach students the skills to join any future research team in academia or business and quickly become a contributing member. It will train them explicitly on topics such as data management, approval processes, funding, that as a researcher they are expected to know but rarely taught.
Education Methods	Lectures, workshops, group work, group study.
Minor Assessment	Assessment includes presentations, posters, writing assignments, oral examination, participation. No exams.
Maximum number of participants	36

NB3510	CSBB part 1: Understanding Questions	15
Responsible Instructor	Dr. H. Mira Bontenbal	
Course Coordinator	J.C. Colgrove	
Instructor	Dr. E. Kalmár	
Instructor	Dr. H.J. Geertsema	
Instructor	M.J.Y. Wackers	
Instructor	D.H.M. Meijer	
Contact Hours / Week x/x/x/x	15/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	None	
Course Contents	We have planned a kick off week for the first week, 8 weeks of work, and a final report symposium in Q1. There are weekly lectures and workshops provided by a variety of experts on four different threads: communication, collaboration, research and reflection (professional skill development). There will be substantial groupwork and self-study. For clarification, the different ECs stated below do not represent independent courses. In Q1 Monday mornings will consist of reflecting on previous week and planning group tasks for the coming week. Lectures on topics around skills development and research. Tuesday is a self/group study day Wednesday: a half day workshop on a professional skills topic, more self-study Thursday: integration of what students have learned so far, writing assignments, discussions with supervisors Friday: reflect on week, symposium with all teams to share what students have learned so far. Communication (~2 ECs) There will be lectures and workshops about technical and advanced literature review, interviewing stakeholders, critical editing/feedback, data visualization, presentations, and students will use those skills in developing their research question. Collaboration (~2 ECs): Workshops and/or lectures within this thread will include teamwork, conflict management and resolution, peer feedback and team building. Research (~9 ECs): The students will work in small groups, building their knowledge, to generate a deep understanding of the biomedical/societal problem they've selected to focus on within Integrative Neuromedicine (topic 2023-2024). They will share their weekly developments with other groups on Friday afternoons through pitches. Other groups will peer review the weekly plans and ideas. Because students from different disciplines will work within the same groups, diverse knowledge and points of view will be shared to provide a transdisciplinary educational experience. They will develop a research question and proposal. There will also be workshops and/or lectures related to this thread, such as perspectives and values, research funding, patents, knowledge utilization, etc Reflection (~2 ECs): Reflective skills are integrated throughout the minor. In workshops, lectures and discussions. This will provide the students with skills and power to reflect on themselves, their personalities, their position within the group, their science and what they expect to gain from the Minor. Students will have reflective writing assignments.	
Study Goals	Q1: At the end of Q1 students are able to: 1. Communicate: a) Create an effective poster presentation. b) Effectively deliver a poster presentation at a research symposium. c) Adapt a poster presentation to various audiences (peers, stakeholders and non-academics). d) Write an effective research proposal to communicate their research plan (background, question, method/approach). e) Apply interview and listening strategies. 2. Collaborate: a) Collaborate effectively with other group members, other groups and the case owners / experts of the field. b) Explain how intercultural differences play a role in collaborationthis includes social (language, nationality, etc.) but also professional (business vs education, medicine vs engineering) differences and apply this knowledge to the collaboration. c) Be aware of and reflect on the various phases of the process, the challenges and problems that were encountered, and what was learned from this. d) Evaluate different roles of members in group projects. e) Provide constructive feedback and show openness to feedback from others. 3. Research: a) Develop a research strategy to learn a body of information. b) Formulate an adequate research question. c) Develop a helicopter view of a research topic and understand the information in different contexts, magnitude and connections. d) Understand the dot on the horizon--Identify interesting/solvable problems, long term possibility and challenges of a research topic. e) Create an overview of the different stakeholders that are involved in the research topic. f) Write an insightful summary of research done with social implications. g) Write a short research proposal with background, knowledge gap, approach and impact section.	

	h) Understand the resources for projects that are available and their costs. i) Able to identify, integrate and analyse current knowledge about a topic including from multiple sources, perspectives and methodologies. 4. Reflect: a) Use framework to reflect on experiences in the minor. b) Ask for constructive feedback of supervisors. c) Understand how personal biases influence your work.
Education Method	Self study, lectures, workshops, group work.
Literature and Study Materials	Course materials will be available in Brightspace and in an online interactive textbook provided via Brightspace.
Assessment	There will be a midterm formative assessment at week 1.5 where students meet with their supervisors and assess their last five-week path. Students will be provided with additional formative assessment and feedback throughout the minor. The final grade consists of assessment of these five products: 30% Report Research proposal (NWO XS proposal format: including literature, research question and methods) GROUP grade 20% Oral exam with supervisor focus on content, includes process as well, interview / oral exam on the proposal - INDIVIDUAL grade 10% Participation grade during weekly pitches, presentations, etc INDIVIDUAL grade 20% Poster presentation to public / stakeholders GROUP grade 20% Critical thinking portfolio: on reaching the learning goals, reflective questions related to lectures and workshops, etc INDIVIDUAL grade Peer feedback on contributions to group work will modify the group grades to achieve individual grades. Because of the nature of the assessments, there are no resits for this course.
Enrolment / Application	Enrollment in Brightspace for the course will be considered registration for all assessments.
Special Information	Students are required to participate in all scheduled course activities. The content of this minor depends on participation and discussion.
Tags	If extenuating circumstances (illness, family emergency) occur, students must contact the minor director before the missed session and develop a plan to address the absence. Adventurous Broad Diverse Ethics Group work Integrated Intensive Project planning / management Small groups

NB3520	CSBB part 2: developing answers	15
Responsible Instructor	Dr. H. Mira Bontenbal	
Course Coordinator	J.C. Colgrove	
Instructor	M.J.Y. Wackers	
Instructor	D.H.M. Meijer	
Instructor	Dr. E. Kalmár	
Instructor	Dr. H.J. Geertsema	
Contact Hours / Week x/x/x/x	0/16/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Q2 Block B 15 ECs: The second half of the Minor will be developing and beginning a research plan or developing a tool/technique/algorithm that can be used to address one or more aspects or solutions to the problem. This second half of the Minor requires a student who has participated in the first half. Education will be a mix of lectures, student-organized events, interviews, workshops, groupwork and self-study. The ECs listed is a guide to workload, not separate courses. Communication (~2 ECs): A series of workshops and/or lectures will be given by experts about how to write grant proposals, grant applications, what the institutional review process is, how to present data analysis and visualize it. Students will also practice these skills in weekly symposiums, and in developing final products. Collaboration (~2 ECs): There will be workshops and/or lectures on how to detect and correct group mistakes, data management, reproducibility. They will also further develop skills by doing active collaboration in developing and doing a project. (Lab) Research (~9 ECs): While the first part of the Minor involves studying the problems raised by the main overarching topic, this second part will delve into researching and finding solutions to these problems. The structure will be similar to the first part. Students will spend most of their time in labs/groups doing research about how to solve the biomedical problems raised during the first part. Workshops and/or lectures include: lab budgets, sustainability in the lab, lab safety, ethics (data privacy storage, integrity), how to use available repositories. Reflection (~2 ECs): The self-reflection skills development will continue in this second part of the Minor but particularly focused on being reflective about the science students are doing. This is core to developing as an ethical scientist. At the end of this block, students will self-reflect on the entire process of this transdisciplinary experience and the Minor. Students will have to write a reflective journal. Topics for workshops and lectures include unconscious bias, critical thinking, troubleshooting.	
Study Goals	Q2: At the end of Q2 students are able to: 1. Communicate: a) Adapt a presentation to various audiences. b) Prepare an effective pitch presentation. c) Effectively deliver a pitch presentation. d) Prepare an effective research presentation / talk . e) Effectively deliver a presentation at a research symposium. f) Write an effective scientific report to communicate their research (relevance, method, findings). g) Provide critical and constructive feedback on scientific reports . 2. Collaborate: a) Develop partnerships with other groups, stakeholders and scientists. b) Reflect on transdisciplinary collaboration in such a way that it can improve collaboration. c) Make a theory-based standpoint in the ethical dilemmas around credit, ownership and group work. 3. Research: a) Develop research plan and gather the right tools and knowledge to complete their project. b) Critically evaluate research strategies, and if necessary, modify and adapt these strategies. c) Plan and (if possible) conduct an experiment based on their research. d) Analyse outcome of an experiment in a critical way. e) Apply research ethics around reproducibility, data management and integrity. f) Understand processes involved with institutional reviews, grant applications and scientific publications. g) Apply ethical and philosophical considerations in performing research. 4. Reflect: a) Use framework to reflect on experiences in minor. b) Ask for constructive feedback from peers. c) Understand how personal biases influence collaboration. Self-study, lab work, lectures, workshops, symposium.	
Education Method		
Assessment	There will be formal formative feedback in week 2.5 based on meeting with supervisor, and informal formative feedback throughout. Final grade will consist of: 30% Report Results paper in the form of a scientific paper GROUP grade 20% Oral exam with supervisor focus on content, includes process as well, interview / oral exam on the results paper-INDIVIDUAL grade	

Enrolment / Application Special Information	10% Participation grade during weekly symposiums, INDIVIDUAL grade 20% Oral presentation to public / stakeholders GROUP grade 20% Critical thinking portfolio: on reaching the learning goals, reflective questions related to lectures and workshops, etc INDIVIDUAL grade Peer feedback on contributions to group work will modify the group grades to achieve individual grades. Because of the nature of the assessments there are no resits. Enrollment in Brightspace is considered registration for all assessments. Students are required to participate in all scheduled course activities. The content of this minor depends on participation and discussion. If extenuating circumstances (illness, family emergency) occur, students must contact the minor director before the missed session and develop a plan to address the absence.
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Year	2023/2024
Organization	Applied Sciences
Education	Minors Applied Sciences

TN-Mi-189-23

Contact for students	Dr. W.G. Bouwman
Program Title	Modern Physics
Director of Education	prof. Chris Kleijn
Program Coordinator	Dr. W.G. Bouwman
In association with the Faculty of	TNW
Introduction 1	Anybody who is not shocked by quantum theory has not understood it (Niels Bohr). Do you have a special talent for Physics and Mathematics, and do you want: to be shocked by the intuitive, unreal world of Quantum Mechanics? to discover what entropy means on a microscopic scale? to gain insight into the properties and measurements of a Higgs particle? to understand the consequences of Einsteins theory of special relativity for space and time? to find out about physical processes in living systems? In that case, the minor in Modern Physics will satisfy your curiosity. The minor covers 20th and 21st-century physics, with courses on Elementary Particle Physics, Quantum Mechanics, Statistical Physics, Thermodynamics and Transport Phenomena in Living Systems, Theory of Relativity, Biophysics and Physics of Radiation Technology. The minor concludes with a group-based Final Experimental Project in one of the Applied Physics departments.
Administration by the Faculty of	TNW
Intended for	Students 3mE, EWI, LR, CiTG, MST (TUD) and International Bachelor Econometrics and Operations Research (EUR), Informatics (UL), Mathematics (UL) and Life, Science and Technology (TUD, UL) under the express condition of having a VWO N&T-diploma
Prerequisites Minor	Vwo-diploma Nature and Technology
Minor Exit Qualifications	Learning Goals 1. Backgrounds of Modern Physics - Quantum mechanics - Statistical Physics - Theory of Relativity - Biophysics - Thermodynamics and Transport Phenomena in Living Systems - Elementary Particle Physics - Physics of Radiation Technology 2. Experimental Physics - setting up, carrying out, analysing and reporting an Experimental Physics Project 3. Literature Research
Education Methods	A mixture of lectures, tutorials, practicals and projects
Minor Assessment	Examination or assignment
Maximum number of participants	50
Minimum number of participants	30

TN2305	Quantum Mechanics for the minor	4
Responsible Instructor	Dr. F. Bociort	
Contact Hours / Week x/x/x/x	0/8/0/0/0/0/0	
Education Period	1B	
Start Education	1B	
Exam Period	1B 2B	
Course Language	English	
Required for	Minor students with basic knowledge of mathematics and physics.	
Expected prior knowledge	The math course from the minor program that you will take in the first term (starting in September) contains important mathematics background that you will need for TN2305. Further you will need basic physics background (high school level, first year university courses in Physics)	
Course Contents	Introduction to quantum mechanics including general concepts and a basic mathematical formalism. We will work on the solutions of the Schroedinger equation in one dimensional potentials such as the harmonic oscillator, square wells, free particle. There will be short presentations of the use of quantum mechanics in modern technology.	
Study Goals	At the end of this course, the student will have an introductory knowledge of basic quantum mechanics that will be enough to follow more advanced quantum mechanics course. This course is equivalent to the bachelor course of the Applied Physics in introduction to quantum mechanics	
Education Method	Two 2-hour lectures per week plus weekly 2 hour exercise classes. Textbook: Introduction to Quantum Mechanics, D.J. Griffiths.	
Literature and Study Materials	Textbook: Griffiths, Introduction to Quantum Mechanics, 2nd or 3rd edition. You may purchase the book at the Delft Physics Student Association VvTP. The international edition is also fine.	
Assessment	Written exam, closed book.	
Elective	Yes	
Tags	Calculus Intensive Involved Physics Quantum	

TN2306	Quantum Engineering and Applications	3
Responsible Instructor	C.K. Andersen	
Contact Hours / Week x/x/x/x	0/0/0/6/0/0/0	
Education Period	2B	
Start Education	2B	
Exam Period	2B 3B	
Course Language	English	
Expected prior knowledge	TN2305	
Course Contents	This course will cover modern topics in applied quantum mechanics and make the students familiar with scientific and engineering aspects of quantum technologies. The topics to be cover are 1) Quantum bits and entanglement - Qubits and quantum circuits - Quantum gates - Bell states 2) Quantum communication - Quantum teleportation - Entanglement swapping - Quantum key distribution 3) Quantum computing - Quantum algorithms - Quantum hardware: Superconducting qubits - Quantum hardware: Spin qubits	
Study Goals	By the end of the course, the students should be able to: - Describe quantum applications for communication and computing - Describe the state of academic and industrial research towards quantum applications - Understand basic protocols for quantum communication - Understand basic algorithms for quantum computing - Calculate the expected outcome of simple quantum applications - Analyse results of simple quantum algorithms	
Education Method	The course will consist of online teaching material, lectures and in-class tutorials. The tutorials will focus on problem solving via analytical calculations or numerical evaluation. There will also be tutorials making use of real quantum hardware in the cloud.	
Assessment	The final examination will be a three hours open-book exam on problems similar to those encountered during the course. The problems at the exam will be either conceptual, analytical or numerical questions.	

TN2612	Theory of Relativity	3
Responsible Instructor	Dr. L. van Eijck	
Contact Hours / Week x/x/x/x	4/0/0/0/0/0/0	
Education Period	1A	
Start Education	1A	
Exam Period	1A 2A	
Course Language	English	
Course Contents	Special Relativity. Einstein's axioms, Lorentz transformations, time dilatation and Lorentz contraction, space-time geometry, Minkovski diagrams, relativistic kinematics and Doppler effect, four-vectors, mass-energy equivalence, relativistic dynamics.	
Study Goals	Getting acquainted with the shortcomings of Newtonian mechanics and with the theory of relativity. Learning to understand the consequences of constant velocity of light and Lorentz transformation, such as length contraction and time dilation and equivalence of mass and energy. Learning the concept of four-vectors.	
Education Method	8 * (1 hour lecture, directly follow by 1 hour excercises)	
Literature and Study Materials	W.D. McComb, 'Dynamics and Relativity', Oxford University Press. Chapters 11, 12, 13, 14.	
Assessment	Written exam	

TN2625	Statistical Physics for the minor	4
Responsible Instructor	Dr. S.R. Parnell	
Contact Hours / Week x/x/x/x	0/0/8/0/0/0/0	
Education Period	2A	
Start Education	2A	
Exam Period	2A 2B	
Course Language	English	
Course Contents	The course is based on the book of Daniel V. Schroeder, Introduction to Thermal Physics, by Addison Wesley Longman ISBN 0 -201-38027-7. ===== Week 1 : First meeting Structure of the course Introduction to Temperature, the Ideal Gas Model and the Equipartition Theorem Probabilities, Microstates-Macrostates, Binomial distribution, Limit of Large Numbers, Stirlings Approximation, Gauss Function Introduction to the models of the Two-State Paramagnet and the Einstein Solid. 2nd and 3rd Laws of Thermodynamics, Application of counting to the ideal gas: thermally or mechanically interacting ideal gases, Extensive and Intensive Physical quantities, Entropy and Temperature ===== Week 2 : Temperature, Heat capacity, Entropy of the ideal gas, Entropy of mixing Two state paramagnet in a magnetic field Negative temperatures. Heat capacity and Phase transitions, Specific Heat of the Einstein Solid, Chemical potential, Chemical Potential of an ideal Gas and of the Einstein Solid Enthalpy, Helmholtz and Gibbs Free Energies, Chemical potential of Mixture of Gases, of Chemical Reactions ===== Week 3 : The Boltzmann Distribution: introduction, the Two State Paramagnet, the Einstein solid. The Partition Function: introduction and examples. Equipartition theorem re-visited. Ideal gases: Maxwell speed distribution and Partition function. Partition Function and Helmholtz free Energy. Quantum Statistics, Gibbs factor ===== Week 4 : Bosons-Fermions, Fermi gas, Blackbody radiation.	
Study Goals	Learning how the statistical approach leads to an understanding of physical phenomena in classical and quantum mechanics.	
Education Method	The course consists of two hours of lectures and two hours of tutorials based on the exercises of book.	
Assessment	The final examination the students are asked to solve problems, which are closely related to the problems dealt during the tutorials and discussed during the contact hours. Besides these problems, a part of the final examination includes conceptual questions, which test the basic understanding of the topic by the student.	

TN2985	Introduction to Methods in Physics	2
Responsible Instructor	Dr. C.F.J. Pols	
Contact Hours / Week x/x/x/x	1A/0/0/0/0/0/0	
Education Period	1A	
Start Education	1A	
Exam Period	none	
Course Language	English	
Course Contents	The course consist of three parts: 1 Data-analysis in Python 2 Experiment 1 3 Experiment 2	
Study Goals	The goal of the course Introduction to Experimentation is to give you a first glimpse of what it means to do experiments in physics, and to demonstrate the rules that must be applied when doing such experiments. You will work with equipment that is often used by physicists in labs. You will learn how to collect, process, analyse and present experimental data. Explicitly, after following this course you are able to: 1. understand what is required before you can start collecting data in a scientific, physics experiment 2. collect, analyse and present empirical data for the purpose of scientific research 3. write a concise report on a physics experiment	
Education Method	Lecture and selfstudy material for data-analysis in Python. On campus experiments	
Reader	A complete manual is available. All relevant information is included.	
Assessment	Online test data-analysis in Python (20%) Scientific report for experiment 1 (40%) Scientific report for experiment 2 (40%)	

TN2986	Mathematical Methods for Modern Physics	2
Responsible Instructor	Prof.dr. B.M. Terhal	
Contact Hours / Week x/x/x/x	6/0/0/0/0/0/0	
Education Period	1A	
Start Education	1A	
Exam Period	1A 2A	
Course Language	English	
Course Contents	This course is a refresher and an introduction to the math necessary to smoothly follow courses on statistical mechanics and quantum mechanics. Topics include: linear algebra, coordinate systems, solving differential equations, Fourier transforms, integrals, differentiation and series.	
Study Goals	The student having completed this course should be aware of the topics mentioned above. That is, he/she should 1. Have knowledge about the theory 2. be able to solve problems within the topics addressed 3. Recognize the equations and know how to deal with them	
Education Method	The course mainly consists of problem solving. Preparation is mainly done using the course material in the form of 9 notebooks which contain exercises. In the lecture, this material will be summarized, explained and example problems will be solved.	
Books	An additional reference book for students who wish to study beyond the notebooks is "Mathematical Methods for Physicists: A comprehensive guide" by Arfken, Weber and Harris (Elsevier). Older versions of this "Mathematical Methods for Physicists" book by Arfken are similarly useful.	
Assessment	The course will be assessed through a single written exam on the topics covered in all nine notebooks.	

TN2994	Experimental and Integrating Final Project	9
Responsible Instructor	Dr. W.G. Bouwman	
Contact Hours / Week x/x/x/x	0/20/20/4/0/0/0/0	
Education Period	1B 2A 2B	
Start Education	1B	
Exam Period	none	
Course Language	English	
Expected prior knowledge	In order to participate, one has to have completed the lab part of TN2985 with a sufficient grade	
Course Contents	Scientific research in a multidisciplinary team and writing an appropriate report on the basis of the formulation of a problem in the field of (applied) physics.	
Study Goals	Introduction to research in (experimental) physics in one of the research groups of the department of Applied Science in which the student shows: -to be able to find and read relevant scientific publications -having or obtaining theoretical knowledge in physics and the ability to apply it; - having adequate experimental skills, such as making an analysis of the defined problem, understanding methods of measurements and measurement instruments, carrying out measurements systematically and effectively, working orderly, making and keeping to the division of tasks (teamwork) and communicating with partners and supervisor. -having the ability to make a proper report of an experiment, such as keeping up a lab journal and (collectively) writing a structured and understandable report - presenting the research for an audience	
Education Method	Project in one of the research groups of Applied Physics; around 112 hours of literature study and writing an literature review, around 70 hours (18 half days) experimental work and an equivalent effort for preparation and reporting.	
Assessment	On the basis of experimental work, reporting and presentation the grade will be composed by the weighted average of literature review (25 %), research proposal (20%), experimental skills (35%), reporting (20%).	

Year	2023/2024
Organization	Applied Sciences
Education	Minors Applied Sciences

TN-Mi-219	
Director of Studies	Dr. M. Blaauboer
Program Coordinator	E.E.S. de Haan
Program Goals	The goal of the minor is to make students familiar with developments in the field of quantum science, quantum technology and quantum devices, and with practical applications arising from research in this multidisciplinary field. This implies a broadening of the knowledge of the students, combined with research experience in this modern field.
Administration by the Faculty of	TNW
Minor Title	Quantum Science and Quantum Information.
EC Minor (Volume)	30 ECTS
Minor Semester	Fall
Cooperating Academic Chair	Department of Quantum Nanoscience QuTech
Contact for Students (Minor)	Eva de Haan, Minor-QSQI-TNW@tudelft.nl
Intended for	Participation in the QSQI minor is open to students from BSc programmes (Applied) Physics, Electrical Engineering, (Applied) Mathematics, and Computer Science at Delft University of Technology and other universities. Students enrolled in the BSc Molecular Science & Technology can (only) participate after successfully completing the Quantum Chemistry and Physics, Linear Algebra and Differential Equations, and Theoretical Chemistry courses of the major Materials. This minor programme requires your full commitment. It should be completed within one academic year.
Expected prior Knowledge	This minor is challenging and is only intended for students with a strong background in calculus and linear algebra. We strongly recommend that you have obtained an average mark for the calculus and algebra courses in your BSc programme that is above 7. This minor should be completed within one academic year.
Minor Exit Qualifications	1. The student possesses knowledge of the field of quantum mechanics and quantum information theory, including the necessary mathematical skills, which enable him/her to participate in basic quantum research and to follow more specialised courses in the field of quantum science and engineering. The student has knowledge of the realisation of quantum devices. 2. The student has knowledge of and experience with research in the field of quantum science, in particular quantum information science. 3. The student knows about the implications of quantum technology and is capable of forming informed opinions on the latter and of defending these. 4. The student has experience with carrying out project work in teams. 5. The student has an overview of the developments in the field of quantum science, technology and information in the recent past and of the prospects for the near future, and is capable of reflecting on these.
Minor Content 1	See the menu on the left for information about the individual course modules.
Education Methods	The minor consists of an integrated set of course modules combined with a final project, in which students work in small groups on the design, hardware and/or software components of a quantum device to apply their newly attained skills. -The number of contact hours is 16-20 hours a week. -Fulltime availability is required for the group project in January. -Timetabling of the course modules is based on four periods in the semester.
Maximum number of participants	60

Year	2023/2024
Organization	Applied Sciences
Education	Minors Applied Sciences

Minor QSQI for AS/TNW students

TN3105	Mathematics for Quantum Physics	2
Responsible Instructor	Dr. M.T. Wimmer	
Instructor	Dr. A.A.F.M. Artaud	
Contact Hours / Week x/x/x/x	6 contact hours per week for three weeks, plus tests	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Required for	minor Quantum Science and Quantum Information	
Expected prior knowledge		
Course Contents	This course is an introduction to the math necessary to smoothly follow courses on quantum mechanics. Topics include: coordinate systems, complex numbers, Fourier transforms, linear algebra including Hilbert spaces, second order differential equations.	
Study Goals	- The student can reproduce elementary formulas from the topics covered. - The student has mathematical problem-solving skills to cope with the follow-up courses of the minor. - The student has knowledge of Hilbert spaces of (in)finite dimension.	
Education Method	In this course, emphasis is on problem-solving skills, in line with the assessment. There are six contact hours during three weeks, of which 1 or 2 hrs are used for explanation in class, the remaining time is spent on solving the problems. Collaborating is stimulated. There is material available on Brightspace which is suitable for self-study. The plenary lecture is only used for providing a global overview of the material.	
Course Relations	This course is one of the introductory courses minor Quantum Science and Quantum Information. The introductory courses will lay the foundation for understanding quantum technology.	
Literature and Study Materials	The course material is available via Brightspace.	
Assessment	The assessment concerns the different subjects within calculus, analysis, and linear algebra addressed in the course. Knowledge and problem-solving skills will be assessed based on the potential of the student to solve problems in these areas. The course will be assessed through three intermediate tests. You can pass the course based on these three tests. If you fail, you need to do a single, 3-hours retake on all topics. The format of the intermediate tests will be communicated at a later stage, depending on the covid situation.	
Permitted Materials during Tests	Pen and paper. A (basic) calculator.	

TN3126	Information and Computation	5
Responsible Instructor	Dr. M. Möller	
Instructor	Dr. J.L.A. Dubbeldam	
Instructor	Dr. R. Ishihara	
Contact Hours / Week x/x/x/x	10 contact hours per week for four weeks, plus tests	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Required for	Physics students participating in the minor Quantum Science and Quantum Information.	
Course Contents	Computer architecture, organization and instructions; Digital systems, coding, switching functions, boolean algebra; Logic gates, gate networks, universality; Sequential systems, Finite state machines; Complexity theory; Information theory; Classical error correction.	
Study Goals	- The student has a basic knowledge of computer architecture. - The student is able to translate non-digital functions to a digital representation. - The student is able to analyze and simplify gate networks. - The student is able to explain combinational systems. - The student is able to explain the behavior of transistors and design basic logic gates. - The student can quantify the information content of a random variable. - The student can derive basic properties of probability distributions and apply them to derive basic properties of entropic quantities. - The student can sketch Shannon's noisy coding theorem and apply it to compute the capacity of a noisy channel. - The student understands the concepts of the minimum distance of a code, can relate it to the error-correcting capabilities of a linear code, and find the minimum distance of small codes. - The student can find the basic properties of block codes, such as their rate and various probabilities of error - Given a source, the student can bound the length of the optimal code and compress the source with explicit codes such as the Huffman code.	
Education Method	Lectures followed by tutorials.	
Course Relations	This course is one of the introductory courses minor Quantum Science and Quantum Information. The introductory course will lay the ground foundation of computer science and information theory necessary for understanding quantum computation.	
Literature and Study Materials	The Feynman lectures on computation contain the material that we will cover at a high level. For the computation part of the course we will follow selected chapters from the book "Digital Design and Computer Architecture" from David Money Harris and Sarah L. Harris, for the information theory part, we will follow selected chapters from "Information Theory, Inference, and Learning" from MacKay, for the physics of computation part we will follow selected chapters from Semiconductor physics and devices. D. Neaman.	
Assessment	The course will be assessed through three intermediate tests and one final exam, the latter on all topics. The final grade will be computed according to the formula 50% average grade of the three intermediate tests and 50% grade of the final exam. The final exam must be passed with a minimal grade of 5.0. There is one additional opportunity that consists on a 3-hours retake on all topics and which will not be averaged with the grades for the three intermediate tests.	
Permitted Materials during Tests		

TN3145	Quantum Communication and Computation	4
Responsible Instructor	T.H. Taminiau	
Contact Hours / Week x/x/x/x	8 contact hours per week for four weeks	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Required for	Minor in Quantum Science and Quantum Information.	
Expected prior knowledge	Linear algebra, quantum physics and its mathematical description.	
Course Contents	The course introduces central topics in quantum communication and quantum computing, including - quantum bits, their representation using state vectors and density matrices, and fidelity with respect to a target qubit - quantum gates and quantum circuit diagrams - the no-cloning theorem and optimal quantum cloning machines - entanglement, its roles as a fundamental property of nature and a resource for quantum technology, and the Bell inequality - quantum key distribution and the impact of eavesdropping - quantum teleportation, entanglement swapping, and quantum repeaters - Shors algorithm	
Study Goals	The student will acquire basic knowledge about the underpinning theoretical aspects of quantum communication and computation, and will learn how to solve simple problems based on quantum mechanical aspects such as superposition and entanglement.	
Education Method	In this course, lectures are combined with group discussions and (supervised) in-class exercises based on literature that is to be studied at home.	
Literature and Study Materials	Lecture notes and other material used during the lectures (to be provided during the course). A good book for additional reading is Michael A. Nielsen and Isaac L. Chuang, Quantum Computation and Quantum Information, Cambridge University Press.	
Assessment	A weekly quiz and a final written exam to assess knowledge and problem solving skills. A formula sheet will not be provided.	
Permitted Materials during Tests	Non-programmable calculator	

TN3156	Quantum Sensing and Measurement	4
Responsible Instructor	Prof.dr. G.A. Steele	
Instructor	T. van der Sar	
Instructor	C.A. Potts	
Contact Hours / Week x/x/x/x	8 contact hours per week for four weeks, plus tests	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Required for	minor Quantum Science and Quantum Information.	
Expected prior knowledge	Mathematics for Quantum Physics, Quantum Physics, Information and Computation. The student should be familiar with the quantisation of the harmonic oscillator, in particular the operator formulation thereof. The student should be familiar with the basic postulates of quantum mechanics and quantum measurement, along with the formalism for predicting the outcomes of quantum measurements using the generalised statistical interpretation (at the level, for example, of chapter 3 of Griffiths).	
Course Contents	- The course will begin with an introduction and review of the (classical) concepts of sensitivity, noise, and power spectral density. - The basics of the harmonic oscillator will be reviewed, including the example of the Harmonic oscillator coherent state and its properties, along with the quadrature representation of classical and quantum coherent fields, and wigner functions - A first example of quantum noise will be introduced using the harmonic oscillator: in particular the concept of quantum (photon) shot noise in coherent states. - The paradigm of continuous measurement will be covered intuitively starting from the generalised statistical interpretation, and then using the Monte Carlo wavefunction technique - These concepts be used to understand the shot noise limit for imprecision noise, and intuitively explore the radiation pressure shot noise limit of optomechanics, a prototypical example of quantum backaction noise. Combining these, we will explore the standard quantum limit (SQL) of sensing and measurement. - The density matrix will be introduced as a tool for combining concepts of quantum and classical noise in qubits and in quantum optics - The mathematics of quantum noise will be formalized using the density matrix and the Lindblad master equation - The QuTiP software package for simulating dynamics of open and monitored quantum systems will be introduced - Basic examples of quantum dynamics will be illustrated using QuTiP - In the last component of the course, the students will simulate a physical process of quantum dynamics combining noise and quantum measurement in a quantum system of their choice, with support from the staff	
Study Goals	The student is able to calculate noise in measurements, calculate the influence of averaging time and measurement bandwidth on measured noise, calculate power spectral densities The student is able to use and explain noise distributions The student is able to calculate the thermal noise spectrum of a harmonic oscillator The student is able to calculate the quantum noise of the ground state of a harmonic oscillator in quadrature representation The student is able to provide an interpretation of the meaning of zero point fluctuations The student is able to perform calculations using the coherent state of the harmonic oscillator The student is able to calculate the shot noise of a coherent state The student is able to reproduce the definition of the Wigner function The student is able to perform calculations with the Wigner function The student is able to provide a visual interpretation of the Wigner function The student is able to interpret the origin of the photon number shot noise of the coherent state using Wigner function visualisation The student is able to identify the Wigner function of squeezed states and Schroedinger cat states The student is able to provide an interpretation of continuous quantum measurement using the Monte Carlo Wave Function Method The student is able to describe the statistics of quantum collapse events under continuous measurement The student is able to write down the density matrix for pure and mixed quantum states The student is able to perform calculations of expectation values using the density matrix The student is able to predict the probability of a measurement outcome using the density matrix The student is able to write down and solve the differential equations for the density matrix elements starting from the Lindblad master equation The student is able to predict the dynamics of the expectation values of observable under continuous quantum measurement using the Lindblad master equation The student is able to construct, solve, visualise, and interpret the quantum dynamics of a continuously monitored quantum system using the QuTiP software package	
Education Method	The course will be based on lectures combined with tutorial sessions. The course will be given 3 times per week, with non-	

Literature and Study Materials	graded homework exercises associated with each lecture. When possible, concepts will be illustrated using computational examples in python.
Assessment	The course will be based on the lecture notes found at: https://qsm.quantumtinkerer.tudelft.nl/ The final grade is made up of 2 parts: 75% Exam 25% Final project The final project is submitted in the last week of the course. The exam will take place during the exam week. The final grade is calculated by scaling the weighted points from the exam and project onto a scale from 0-10, rounding any final grades below 1 up to 1.
Permitted Materials during Tests	

TN3166	Solid-State Quantum Bits	9
Responsible Instructor	Dr.ir. M. Veldhorst	
Contact Hours / Week x/x/x/x	2 * 8 contact hours per week for four weeks	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	minor Quantum Science and Quantum Information.	
Expected prior knowledge	Mathematics for Quantum Physics, Quantum Physics, Information and Computation.	
Course Contents	This course comprises two subjects: Semiconductor Quantum Bits and Superconductor Quantum Bits. The individual parts have their own entries in the study guide and Brightspace, under the codes TN3166-A and TN3166-B. We refer to these entries for detailed information on the contents.	
Study Goals	The student has knowledge about Semiconductor Quantum Bits and Superconductor Quantum Bits. See the individual entries in the study guide for details.	
Education Method	See the individual entries in the study guide for each part: Semiconductor Quantum Bits and Superconductor Quantum Bits.	
Assessment	The final grade will be the weighted average (4/9 and 5/9) of the grades of the two subjects with the proviso that each partial grade must be 5.0 or higher. See the individual entries in the study guide for further details.	

TN3166-A	Semiconductor Quantum Bits	4
Responsible Instructor	Dr.ir. M. Veldhorst	
Instructor	Dr. G. Scappucci	
Contact Hours / Week x/x/x/x	8 contact hours per week for four weeks	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This module is part of TN3166, Solid-State Quantum Bits. A valid final grade and credits are only obtained for the TN3166 course and not for this module on its own.	
Expected prior knowledge	Mathematics for Quantum Physics, Quantum Physics, Information and Computation.	
Course Contents	Semiconductor quantum bits represent a promising platform for practical quantum information. In this course, you will become familiar with semiconductor devices. Questions as how do they look like, how can we execute quantum operations with these devices, and what are the challenges and opportunities with these systems, are addressed in this course. The syllabus includes two-level systems, the Rabi model, Bloch sphere representation, quantum operations, universal gate set, energy bands, energy scales, semiconductor heterostructures and devices, quantum dots, donors and defects, exchange between spins, and scaling up qubits.	
Study Goals	After the course, the student should be able to - list and explain the relevant criteria for quantum computation - explain how materials for semiconductor quantum devices are made, illustrate how structural properties are inspected, and evaluate the electrical properties - solve elementary problems in the definition and operation of quantum dots and illustrate the impact of noise and strategies to mitigate certain noise - assess the challenges and opportunities of semiconductor quantum bits and design quantum dot qubits capable of executing universal quantum logic.	
Education Method	In this course, weekly lectures are combined with online material and exercises based on literature on the subject of semiconductor qubits. Additionally, small group projects will be carried out where students jointly analyze, assess, and create.	
Literature and Study Materials	Lecture notes, presentations slides and online video and lectures. In addition scientific and review articles will be used.	
Assessment	Final grade will be based on assignments (30%) and a final written exam (70%). Knowledge and problem solving skills will be assessed based on the potential of the student to solve problems in these areas.	
Permitted Materials during Tests	Closed book exam. Pen and calculator are permitted.	

TN3166-B	Superconductor Quantum Bits	5
Responsible Instructor	C.K. Andersen	
Instructor	Dr. L. di Carlo	
Contact Hours / Week x/x/x/x	4-8 contact hours per week for six weeks	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This module is part of TN3166, Solid-State Quantum Bits. A valid final grade and credits are only obtained for the TN3166 course and not for this module on its own.	
Expected prior knowledge	Mathematics for Quantum Physics, Quantum Physics, Information and Computation.	
Course Contents	Phenomenological theory of superconductors, Josephson junctions, devices, quantum circuits and cavities.	
Study Goals	- The student can describe superconductivity on the phenomenological level and can list their material parameters, and can reproduce the equations of the Josephson effect. - The student can perform calculations with the quantized harmonic oscillator, and with circuits including Josephson elements. - The student can distinguish between and describe various superconducting quantum bits.	
Education Method	In this course, emphasis is on understanding the basic of superconducting circuits and problem solving skills. Collaborating is stimulated. The lecture slides, assignments and quizzes will form the content of the course.	
Literature and Study Materials		
Assessment	Final grade will be based on assignments (30%) and a final written exam (70%). Knowledge and problem solving skills will be assessed based on the potential of the student to solve problems in these areas. A formula sheet will not be provided in order to test knowledge of elementary formulas.	
Permitted Materials during Tests		

TN3175	Quantum Engineering Group Project	6
Responsible Instructor	Dr.ir. M. Veldhorst	
Instructor	Dr. G. Scappucci	
Contact Hours / Week x/x/x/x	Full time from Monday January 3rd to Wednesday January 26th.	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	minor Quantum Science and Quantum Information.	
Expected prior knowledge	All preceding courses of the minor Quantum Science and Quantum Information. Passing at least 16 ec of those courses is required to participate in this project.	
Course Contents	Students work in small groups (4-5 students) in a project on a quantum science and information topic, and actively interaction with a quantum computer in the cloud (real or simulator). Introductory lectures are given to acquaint students to program and work with quantum simulators and real quantum systems.	
Study Goals	After the course, student should be able to - program and execute small rudimentary quantum algorithms on a real quantum processor or simulator - explain the components relevant in a quantum computer stack, including the software and hardware elements. - work effectively in a group, and communicate in a positive and constructive manner - present clearly the results by means of an oral presentation and a written report.	
Education Method	Project team work (4-5 students, mixed educational background) supervised by TAs and staff, with intermediate reporting and presentations.	
Course Relations	In this project, the skills and knowledge of the previous courses of the minor Quantum Science and Quantum Information come together.	
Literature and Study Materials	Will be provided and students are activated to search for the literature needed to do their project	
Assessment	The assessment is based on project planning (10%), project execution (50%), written report (25%) and presentation (15%). A peer review is conducted as part of the final assessment and used to establish individual marks.	
Enrolment / Application	Passing at least 15 ec of courses of the Quantum Science and Quantum Information minor is required to be admitted to the group project.	

Year	2023/2024
Organization	Applied Sciences
Education	Minors Applied Sciences

Minor QSQI for EEMCS/EWI students

TN3105	Mathematics for Quantum Physics	2
Responsible Instructor	Dr. M.T. Wimmer	
Instructor	Dr. A.A.F.M. Artaud	
Contact Hours / Week x/x/x/x	6 contact hours per week for three weeks, plus tests	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Required for	minor Quantum Science and Quantum Information	
Expected prior knowledge		
Course Contents	This course is an introduction to the math necessary to smoothly follow courses on quantum mechanics. Topics include: coordinate systems, complex numbers, Fourier transforms, linear algebra including Hilbert spaces, second order differential equations.	
Study Goals	- The student can reproduce elementary formulas from the topics covered. - The student has mathematical problem-solving skills to cope with the follow-up courses of the minor. - The student has knowledge of Hilbert spaces of (in)finite dimension.	
Education Method	In this course, emphasis is on problem-solving skills, in line with the assessment. There are six contact hours during three weeks, of which 1 or 2 hrs are used for explanation in class, the remaining time is spent on solving the problems. Collaborating is stimulated. There is material available on Brightspace which is suitable for self-study. The plenary lecture is only used for providing a global overview of the material.	
Course Relations	This course is one of the introductory courses minor Quantum Science and Quantum Information. The introductory courses will lay the foundation for understanding quantum technology.	
Literature and Study Materials	The course material is available via Brightspace.	
Assessment	The assessment concerns the different subjects within calculus, analysis, and linear algebra addressed in the course. Knowledge and problem-solving skills will be assessed based on the potential of the student to solve problems in these areas. The course will be assessed through three intermediate tests. You can pass the course based on these three tests. If you fail, you need to do a single, 3-hours retake on all topics. The format of the intermediate tests will be communicated at a later stage, depending on the covid situation.	
Permitted Materials during Tests	Pen and paper. A (basic) calculator.	

TN3136	Quantum Physics	5
Responsible Instructor	V.V. Dobrovitski	
Instructor	Prof.dr. Y.M. Blanter	
Contact Hours / Week x/x/x/x	8 contact hours per week for five weeks, plus tests	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Required for	non-Physics students participating in the minor Quantum Science and Quantum Information.	
Course Contents	Elementary quantum mechanics; Schrödinger equation, simple quantum mechanical systems, angular momentum, spin, many-particles/many-spin systems.	
Study Goals	- The student has knowledge about the fundamental concepts of quantum mechanics (wave-particle duality, wave function, operators and observables, tunnelling, quantum measurement etc), and can apply these concepts to simple problems. - The student can work with the Schrodinger equation, solve simple one-dimensional quantized systems, and interpret results in terms of the statistics of outcomes. - The student understands the concept of angular momentum and spin. Can solve simple problems about spin 1/2, interpret it as a two-level system. - The student understands and can construct the basic notions for theoretical description of a few-spin system (wave function, basic operators and observables).	
Education Method	Lectures followed by seminars/tutorials, including supervised problem solving, and independent reading and problem solving.	
Course Relations	This course is one of the introductory courses minor Quantum Science and Quantum Information. The introductory courses will lay the ground foundation for understanding quantum technology.	
Literature and Study Materials	No mandatory textbooks. Suggested optional reading: lecture notes and slides, books "Introduction to Quantum mechanics" by A. C. Philips and "Introduction to Quantum Mechanics" by D. J. Griffiths.	
Assessment	The students take four tests during the course, with the final grade being an average of the grades for the tests. The homework is not mandatory, but can be taken into account with the weight of 15% if it improves the final grade (in this case the final grade will be determined as $0.15 \times [\text{average homework grade}] + 0.85 \times [\text{average test grade}]$). If the student wants to improve the final grade, then the student can take a final exam over the content of the whole course, and the final grade will be a maximum of two grades (the average grade over the tests, and the grade for the final exam).	
Permitted Materials during Tests	The tests (and the resit exam) are closed-book supervised written tests/exams. Hand-written notes from lectures and seminars, written by the student personally, are allowed, but no copies, no printouts, and no problem solutions. The use of calculators or cell phones is not allowed during the exam.	

TN3145	Quantum Communication and Computation	4
Responsible Instructor	T.H. Taminiau	
Contact Hours / Week x/x/x/x	8 contact hours per week for four weeks	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Required for	Minor in Quantum Science and Quantum Information.	
Expected prior knowledge	Linear algebra, quantum physics and its mathematical description.	
Course Contents	The course introduces central topics in quantum communication and quantum computing, including - quantum bits, their representation using state vectors and density matrices, and fidelity with respect to a target qubit - quantum gates and quantum circuit diagrams - the no-cloning theorem and optimal quantum cloning machines - entanglement, its roles as a fundamental property of nature and a resource for quantum technology, and the Bell inequality - quantum key distribution and the impact of eavesdropping - quantum teleportation, entanglement swapping, and quantum repeaters - Shors algorithm	
Study Goals	The student will acquire basic knowledge about the underpinning theoretical aspects of quantum communication and computation, and will learn how to solve simple problems based on quantum mechanical aspects such as superposition and entanglement.	
Education Method	In this course, lectures are combined with group discussions and (supervised) in-class exercises based on literature that is to be studied at home.	
Literature and Study Materials	Lecture notes and other material used during the lectures (to be provided during the course). A good book for additional reading is Michael A. Nielsen and Isaac L. Chuang, Quantum Computation and Quantum Information, Cambridge University Press.	
Assessment	A weekly quiz and a final written exam to assess knowledge and problem solving skills. A formula sheet will not be provided.	
Permitted Materials during Tests	Non-programmable calculator	

TN3156	Quantum Sensing and Measurement	4
Responsible Instructor	Prof.dr. G.A. Steele	
Instructor	T. van der Sar	
Instructor	C.A. Potts	
Contact Hours / Week x/x/x/x	8 contact hours per week for four weeks, plus tests	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Required for	minor Quantum Science and Quantum Information.	
Expected prior knowledge	Mathematics for Quantum Physics, Quantum Physics, Information and Computation. The student should be familiar with the quantisation of the harmonic oscillator, in particular the operator formulation thereof. The student should be familiar with the basic postulates of quantum mechanics and quantum measurement, along with the formalism for predicting the outcomes of quantum measurements using the generalised statistical interpretation (at the level, for example, of chapter 3 of Griffiths).	
Course Contents	- The course will begin with an introduction and review of the (classical) concepts of sensitivity, noise, and power spectral density. - The basics of the harmonic oscillator will be reviewed, including the example of the Harmonic oscillator coherent state and its properties, along with the quadrature representation of classical and quantum coherent fields, and wigner functions - A first example of quantum noise will be introduced using the harmonic oscillator: in particular the concept of quantum (photon) shot noise in coherent states. - The paradigm of continuous measurement will be covered intuitively starting from the generalised statistical interpretation, and then using the Monte Carlo wavefunction technique - These concepts be used to understand the shot noise limit for imprecision noise, and intuitively explore the radiation pressure shot noise limit of optomechanics, a prototypical example of quantum backaction noise. Combining these, we will explore the standard quantum limit (SQL) of sensing and measurement. - The density matrix will be introduced as a tool for combining concepts of quantum and classical noise in qubits and in quantum optics - The mathematics of quantum noise will be formalized using the density matrix and the Lindblad master equation - The QuTiP software package for simulating dynamics of open and monitored quantum systems will be introduced - Basic examples of quantum dynamics will be illustrated using QuTiP - In the last component of the course, the students will simulate a physical process of quantum dynamics combining noise and quantum measurement in a quantum system of their choice, with support from the staff	
Study Goals	The student is able to calculate noise in measurements, calculate the influence of averaging time and measurement bandwidth on measured noise, calculate power spectral densities The student is able to use and explain noise distributions The student is able to calculate the thermal noise spectrum of a harmonic oscillator The student is able to calculate the quantum noise of the ground state of a harmonic oscillator in quadrature representation The student is able to provide an interpretation of the meaning of zero point fluctuations The student is able to perform calculations using the coherent state of the harmonic oscillator The student is able to calculate the shot noise of a coherent state The student is able to reproduce the definition of the Wigner function The student is able to perform calculations with the Wigner function The student is able to provide a visual interpretation of the Wigner function The student is able to interpret the origin of the photon number shot noise of the coherent state using Wigner function visualisation The student is able to identify the Wigner function of squeezed states and Schroedinger cat states The student is able to provide an interpretation of continuous quantum measurement using the Monte Carlo Wave Function Method The student is able to describe the statistics of quantum collapse events under continuous measurement The student is able to write down the density matrix for pure and mixed quantum states The student is able to perform calculations of expectation values using the density matrix The student is able to predict the probability of a measurement outcome using the density matrix The student is able to write down and solve the differential equations for the density matrix elements starting from the Lindblad master equation The student is able to predict the dynamics of the expectation values of observable under continuous quantum measurement using the Lindblad master equation The student is able to construct, solve, visualise, and interpret the quantum dynamics of a continuously monitored quantum system using the QuTiP software package	
Education Method	The course will be based on lectures combined with tutorial sessions. The course will be given 3 times per week, with non-	

Literature and Study Materials	graded homework exercises associated with each lecture. When possible, concepts will be illustrated using computational examples in python.
Assessment	The course will be based on the lecture notes found at: https://qsm.quantumtinkerer.tudelft.nl/ The final grade is made up of 2 parts: 75% Exam 25% Final project The final project is submitted in the last week of the course. The exam will take place during the exam week. The final grade is calculated by scaling the weighted points from the exam and project onto a scale from 0-10, rounding any final grades below 1 up to 1.
Permitted Materials during Tests	

TN3166	Solid-State Quantum Bits	9
Responsible Instructor	Dr.ir. M. Veldhorst	
Contact Hours / Week x/x/x/x	2 * 8 contact hours per week for four weeks	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	minor Quantum Science and Quantum Information.	
Expected prior knowledge	Mathematics for Quantum Physics, Quantum Physics, Information and Computation.	
Course Contents	This course comprises two subjects: Semiconductor Quantum Bits and Superconductor Quantum Bits. The individual parts have their own entries in the study guide and Brightspace, under the codes TN3166-A and TN3166-B. We refer to these entries for detailed information on the contents.	
Study Goals	The student has knowledge about Semiconductor Quantum Bits and Superconductor Quantum Bits. See the individual entries in the study guide for details.	
Education Method	See the individual entries in the study guide for each part: Semiconductor Quantum Bits and Superconductor Quantum Bits.	
Assessment	The final grade will be the weighted average (4/9 and 5/9) of the grades of the two subjects with the proviso that each partial grade must be 5.0 or higher. See the individual entries in the study guide for further details.	

TN3175	Quantum Engineering Group Project	6
Responsible Instructor	Dr.ir. M. Veldhorst	
Instructor	Dr. G. Scappucci	
Contact Hours / Week x/x/x/x	Full time from Monday January 3rd to Wednesday January 26th.	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	minor Quantum Science and Quantum Information.	
Expected prior knowledge	All preceding courses of the minor Quantum Science and Quantum Information. Passing at least 16 ec of those courses is required to participate in this project.	
Course Contents	Students work in small groups (4-5 students) in a project on a quantum science and information topic, and actively interaction with a quantum computer in the cloud (real or simulator). Introductory lectures are given to acquaint students to program and work with quantum simulators and real quantum systems.	
Study Goals	After the course, student should be able to - program and execute small rudimentary quantum algorithms on a real quantum processor or simulator - explain the components relevant in a quantum computer stack, including the software and hardware elements. - work effectively in a group, and communicate in a positive and constructive manner - present clearly the results by means of an oral presentation and a written report.	
Education Method	Project team work (4-5 students, mixed educational background) supervised by TAs and staff, with intermediate reporting and presentations.	
Course Relations	In this project, the skills and knowledge of the previous courses of the minor Quantum Science and Quantum Information come together.	
Literature and Study Materials	Will be provided and students are activated to search for the literature needed to do their project	
Assessment	The assessment is based on project planning (10%), project execution (50%), written report (25%) and presentation (15%). A peer review is conducted as part of the final assessment and used to establish individual marks.	
Enrolment / Application	Passing at least 15 ec of courses of the Quantum Science and Quantum Information minor is required to be admitted to the group project.	

Year	2023/2024
Organization	Applied Sciences
Education	Minors Applied Sciences

TN-Mi-227-21	
Program Structure 1	See: https://www.universiteitleiden.nl/en/science/chemistry/education/minor

Year	2023/2024
Organization	Applied Sciences
Education	Minors Applied Sciences

AP-Mi-181 2019

Program Coordinator	Ir. A.J.W. Haket
In association with the Faculty of	TNW
Options	See http://ap.msc.studyguide.tudelft.nl (under 'special programmes' in the menu on the left side) for information on MSc Applied Physics bridging programmes.
Minor Remarks / Schedule	Note that the bridging programmes can not be completed in one semester.

Year	2023/2024
Organization	Applied Sciences
Education	Minors Applied Sciences

CHE-MI-182 2019

In association with the Faculty of	TNW
Options	See http://chem.msc.studyguide.tudelft.nl (under 'special programmes' in the menu on the left side) for information on MSc Chemical Engineering bridging programmes.
Minor Remarks / Schedule	Note that the bridging programmes can not be completed in one semester.

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